Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

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Code, tests & data

Estimate position bias from ordinary click logs — weight each repeated query–item pair by $\omega_i = \frac{N_k^i \, N_{k'}^i}{N_k^i + N_{k'}^i}$ (harmonic; N = the pair's impressions at each rank).

On the full KDD log, equal weighting needs ≈ 4.5× **more impressions** for the same estimated precision.

(A model-based precision gain computed on the KDD log; independent ground-truth checks in block 6.)

1. Same item, different rank ⇒ position bias

Users click top results more — partly because they are *better*, partly just because they are *seen* more. Under the **Position-Based Model**, a click factorises as

$$P(\text{click} \mid q, d, k) = \underbrace{p_k}_{\substack{\text{examination:} \\ \text{seen at rank } k}} \times \underbrace{r_{q,d}}_{\text{relevance}}.$$

The **same product X**, shown at **different ranks** across groups of logged sessions, lets us cancel relevance:

Sessions: X at rank 1

Query: red shoes
1. Product X
2. other item
3. other item

1,000 impressions → 100 clicks → **10% CTR**

Sessions: X at rank 3

Query: red shoes
1. other item
2. other item
3. Product X

1,000 impressions → 40 clicks → **4% CTR**

Same query & item ⇒ same r, and $\text{CTR}_k = p_k r$:

$$\frac{\text{CTR@1}}{\text{CTR@3}} = \frac{10\%}{4\%} = \frac{p_1 r}{p_3 r} = \frac{p_1}{p_3} = \mathbf{2.5\times}$$

The CTR ratio *is* the position-bias ratio — no A/B test, no relevance labels. Reusing these natural rank changes is called *intervention harvesting*; the task is to combine **millions** of such query–item cells into one estimate of $R_{k,k'} = p_k / p_{k'}$.

Key terms: *cell* = one query–item pair · N_k = times shown at rank k · p_k = chance rank k is examined · ω_i = that cell's weight in the average.

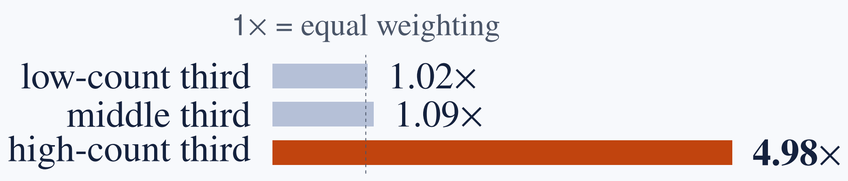
2. Why equal weighting fails

On the **full KDD Cup 2012** log (235.6M impressions), the cells seen at both rank 1 and rank 2 — **4.61M** query–item pairs, 132.3M impressions — have wildly uneven counts:

60.7%
of pairs are near-empty ($N_1 + N_2 \le 5$)

42%
of all impressions sit in the top **0.1%** of pairs

Giving a near-empty cell the **same say** as a million-impression cell is like averaging a noisy guess and a precise measurement equally. The gain from fixing this lives entirely in the high-count cells:



Efficiency vs equal weighting, by cell-count third (KDD). Efficiency = variance under equal weighting ÷ variance under harmonic; higher is better.

3. The one-line fix: a count-only weight

Under PBM, *any* weight that depends only on counts still estimates the same ratio $p_k / p_{k'}$ — it changes only the noise — so we may pick it purely for precision. We compare three:

Scheme	Weight ω_i
Equal *(prior work)*	1
Min-count *(new)*	$\min(N_k^i, N_{k'}^i)$
Harmonic *(new)*	$\frac{N_k^i N_{k'}^i}{N_k^i + N_{k'}^i}$

The entire change, per query–item pair — replace the weight, nothing else:

$$\omega_i = 1 \longrightarrow \omega_i = N_k^i \, N_{k'}^i \, / \, (N_k^i + N_{k'}^i)$$

One line. No experiment, no relevance labels, no tuned parameters.

4. Result: ranks 1 vs 2 (4.61M pairs, 132.3M impr.)

Scheme	Est. p_1/p_2	SE (lower better)	**Efficiency vs equal**
Equal	1.683	4.24e-3	1.00
Min	1.712	2.05e-3	4.41
Harmonic	**1.710**	**2.02e-3**	**4.53**

4.53×

equal weighting needs ~4.5× the impressions for the same precision

A fair caveat, read with the number: this is *precision* at a fixed target. When PBM is imperfect, different weights average slightly different things (equal → 1.68, harmonic → 1.71) — we do **not** claim one is the single "true" bias. "Efficiency" = Var(equal)/Var(harmonic).

The surprise: the "optimal" oracle is *worse*

Classical theory says: weight by the exact per-cell variance, which needs the relevance r_i. Plug in an *estimated* $\hat{r}_i$ and efficiency **collapses to 0.002** — i.e. ~500× *worse* than equal weighting — because $\hat{r}_i$ is too noisy to be useful on the 60% near-empty cells.

So the count-only recipe isn't a shortcut — it is the robust choice.

5. Why it works

✓ Harmonic is the **inverse-variance optimum** of a count-only approximation to the per-cell variance (the *surrogate* in our title) — dropping the unknown relevance term is exactly what makes it *deployable*.

✓ **Provably no worse** than equal weighting on that surrogate variance, for *any* counts (Cauchy–Schwarz). On KDD's empirical asymmetric surrogate it still improves 4.75×.

✓ **No relevance estimates, no tuning** — min-count has no such guarantee, so we recommend harmonic.

6. Validation & guardrail

Three independent lines of evidence — one *model-based* (a variance formula applied to real data) and two *ground-truth* (true variance known by simulation):

4.5×
KDD real log
model-based asymptotic efficiency

up to 54%
Synthetic ground truth
43.6–54% less variance

~40%
Yahoo semi-synthetic
less variance
(95% CI: 29–51% less)

Count-only weights pull further ahead of equal weighting as the log gets more heavy-tailed — exactly the production regime.

Guardrail. PBM is only an approximation: across 16,132 high-information pairs, the per-cell bias ratios vary ≈3.8× more than click noise alone would explain (Cochran Q). So choosing a weight trades precision against *which* average you estimate — there is no single "true" ratio to find.

Bottom line

Swap equal weighting for the **harmonic count-only weight**: **one line of code**, no new data, ~4.5× more efficient on the full KDD log (ranks 1 vs 2) — and *don't* reach for the plug-in relevance oracle, which is worse than doing nothing on heavy-tailed logs.

github.com/alemagnani/surrogate-ivw-intervention-harvesting • DOI 10.1145/3799682.3839883
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